Interaction and Evidence Module

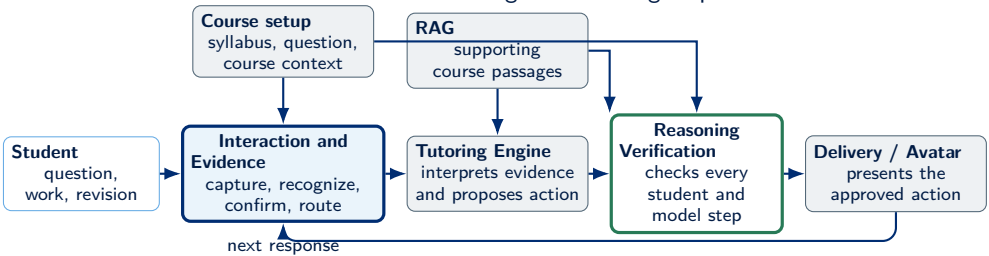
A specification for capturing student work and coordinating a tutoring turn

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Interaction's role in the system

Goal: capture student work without changing its meaning, confirm consequential ambiguity, and pass factual evidence through the tutoring loop.



Interaction owns the blue box only. It does not ingest course content, retrieve passages, diagnose learning, choose or verify tutoring actions, or control avatar presentation.

Spec: `architecture.md` §§3–5; `student-interaction.md` §§1–2

V1 scope and ownership boundary

V1 student inputs

- Typed text, Markdown, code, and TeX
- Handwritten image/PDF with a preserved original and logged resolution/contrast enhancement
- Direct stylus input through a sketchpad
- One recorded voice clip with editable transcript

Interaction owns

Capture, recognition routing, focused confirmation, Student Evidence construction, attempt links, and handoff status.

Other modules own

Course ingestion, RAG ranking, learner diagnosis, tutoring-action selection, reasoning verification, and presentation.

Deferred

Continuous voice, live video, and real-time avatar conversation.

Design rule: coordinate the turn without duplicating another module's decision.

Spec: `student-interaction.md` §§2–4; `architecture.md` §4

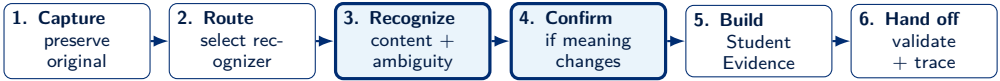
Inputs: what arrives and from where

Input	Producing module	Interaction uses
Student submission	Student/client	Original work, input type, explanation, and stated uncertainty.
Course/question references	Course Context/Ingestion	IDs, display-safe prompt, allowed response modes, and constraints.
Recognition response	Interaction recognizer	Transcription, alternatives, ambiguous regions, model version, and status.
Tutoring Action + verification	Tutoring Engine + Reasoning Verification	Approved action, per-step verification status, and expected response; forwarded unchanged.
Delivery Result	Delivery Layer/Avatar	Presented content, channel, available response modes, and delivery status.
Logging configuration	System/session configuration	Logging, retention, redaction, and cost settings.

Restricted: expected solutions and diagnostic cues never enter recognition or evidence construction.

Spec: `student-interaction.md` §5 and §14; `interfaces.md` §§2–5, 11–12

Expected behavior: one processing contract



Submission	Recognition	Evidence
The original student input and stable course, question, session, turn, and attempt links.	The interpretation, alternatives, uncertainty, and corrections. For handwriting below 300 DPI or 2000 px on its long edge, create a derived 300-DPI/2× copy, then deskew, denoise, and enhance contrast/sharpness. Log every transform; never overwrite the original.	Confirmed factual work and history, without correctness, mastery, or misconception labels.

Spec: `student-interaction.md` §§6–10, 12 and 16; `interfaces.md` §§4–7

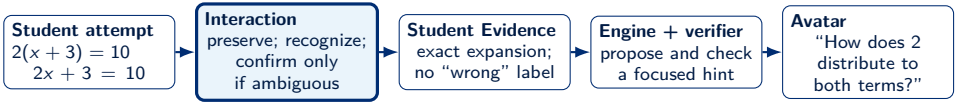
Outputs: what leaves and where it goes

Output	Receiving module	Contract
`StudentSubmission`	Recognizer, evidence builder, session state	Original input with a stable ID and input type.
Recognition result/correction	Delivery confirmation UI; evidence builder	Raw interpretation stays separate from confirmed content.
`StudentEvidence`	Learner State/Tutoring Engine	Validated observable facts and history; no diagnosis.
Retrieval request	RAG/Course Retrieval	Minimum question/evidence context under the agreed routing contract.
Trace event	Session state/diagnostics	Handoff status, record versions, timing, and optional cost.
Confirmation or retry request	Delivery Layer/Avatar	Focused request for information needed to continue.

Every output identifies its producer, consumer, schema version, status, and source records.

Spec: `student-interaction.md` §6 and §14; `interfaces.md` §§4–7, 9–13

Generic example: one tutoring turn



Context received before the attempt

- Active course and question identifiers
- Prompt: solve $2(x+3)=10$
- Expected response mode
- Display-safe concept and constraint references

Next response

The student revises to $2x+6=10$, then $x=2$. Interaction creates new evidence linked to the earlier attempt and the hint; neither attempt is overwritten.

Visible progress: one trace connects context, original work, assistance, delivery, and revision across module boundaries.

Spec: `student-interaction.md` §§8, 14, 16–17; `architecture.md` §5

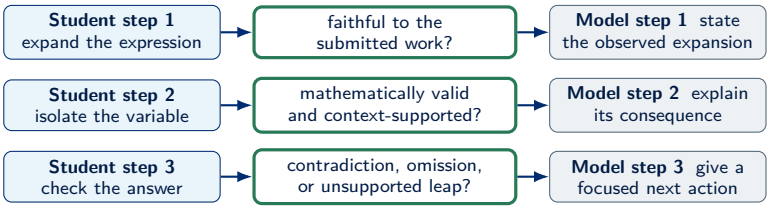
Ambiguity and failure behavior

Condition	Status	Required response
x_2 versus x^2, or another meaning-changing uncertainty	`unconfirmed`	Show the relevant region and candidates; require student edit or confirmation.
No usable content after enhancement, or unsupported input	`invalid/failed`	Preserve according to policy, request a clearer rescan or supported retry, and create no usable evidence.
Only part of the work is recognized	`partial`	Mark missing regions; continue only if downstream policy accepts partial evidence.
Course/question reference is absent	`context_missing`	Preserve the submission but stop the tutoring handoff until repaired.
Reasoning is rejected or cannot be checked	Verification blocked	Return to the Tutoring Engine or retry verification; do not deliver as approved.
Delivery or another handoff fails	Recoverable failure	Retry or request an allowed alternative without changing the tutoring decision.

Recognition never silently repairs a student's mathematical mistake.

Spec: `student-interaction.md` §§10–11, 13–16; `README.md` §7

Verify every student and model step



Pass only when

Every student step is represented accurately, every model step has support, and the response remains consistent with the question and course passages.

Otherwise

Return the failed step, reason, and supporting references to the Tutoring Engine for revision. Unchecked or failed output is not delivered as approved.

The verification layer checks reasoning; it does not choose pedagogy or rewrite student work.

Capture and evidence targets

Area	Measure	Initial target
Text/math recognition	Normalized equation exact match; symbol error rate (SER)	Exact $\geq 90\%$; SER $\leq 5\%$
Recorded voice	Word error rate on the confirmed transcript	WER $\leq 10\%$
Handwriting enhancement	Original retention; logged transforms; low-resolution exact-match gain; enhancement-induced symbol error	100% retained/logged; gain ≥ 10 percentage points; induced error $\leq 1\%$
Evidence integrity	Required ID/link validity; raw/corrected separation; overwritten originals	100% valid/separate; zero overwrites
Ambiguity handling	Recall and precision on meaning-changing ambiguity; confirmations per submission	Recall $\geq 95\%$; precision $\geq 80\%$; median ≤ 1 confirmation

How to calculate recognition quality

Exact match compares normalized expressions. SER = symbol edit distance divided by reference-symbol count. Enhancement gain = post-enhancement exact-match rate minus the unenhanced rate on the same low-resolution pages.

Spec: `student-interaction.md` §§9–11 and 15–17

Reasoning and operation targets

Area	Measure	Initial target
Student-work coverage	Recognized student steps represented in the verification trace	100% of steps accounted for
Model-step validity	Mathematically valid model steps divided by all proposed model steps	$\geq 95\%$
Grounding and safety	Supported-claim precision; false-accept rate; contradiction rate	Grounding $\geq 95\%$; false accepts $\leq 2\%$; contradictions $\leq 1\%$
Turn completion	Tutoring turns ending in a valid Delivery Result or structured recoverable status	$\geq 95\%$
Latency and accounting	End-to-end p95 latency; turns with latency and cost recorded	Typed ≤ 8s; handwriting ≤ 25s; accounting coverage $\geq 99\%$

Tail-risk target

Define turn loss on $[0, 1]$ from math error, unsupported claims, unresolved ambiguity, and failed handoffs. Target $\mathrm{CVaR}_{95} \leq 0.20$, where CVaR_{95} is mean loss in the worst 5% of turns. Recalibrate after the first baseline.

Spec: `student-interaction.md` §§15–17; `README.md` §§7, 10

This week's progress and next decisions

Made concrete this week

- V1 input boundary and ownership
- Named producers and consumers
- Submission, recognition, and evidence contracts
- Ambiguity and failure behavior
- Generic end-to-end tutoring trace
- Proposed reasoning-verification gate

Still open

- Minimum evidence accepted by the engine
- Timeout and retry behavior between modules

These are the only remaining interface decisions identified for this module.

Next milestone: implement one typed tutoring loop, then add handwriting, stylus, and recorded voice through the same contracts.